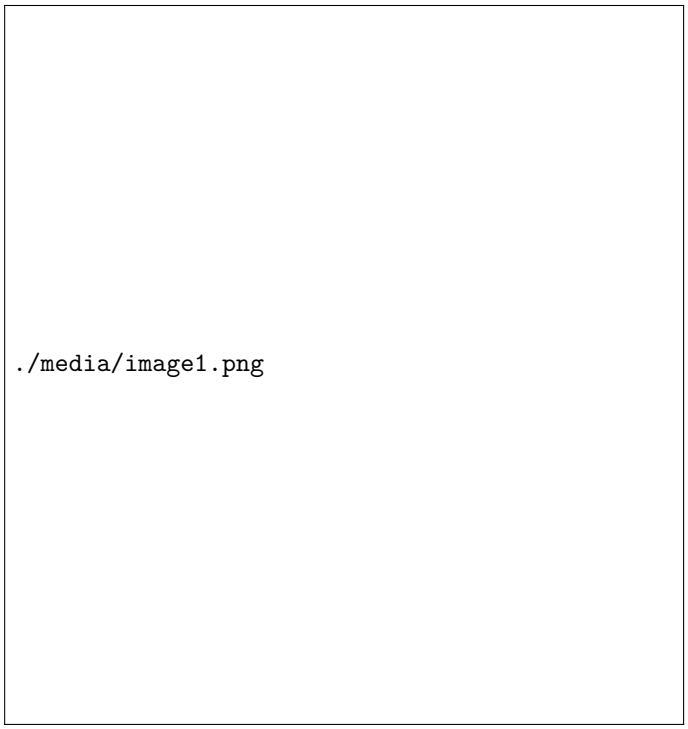
A project submitted in partial fulfillment of the degree of

BS in Software Engineering



Department of Software Engineering

Faculty of Computer Science & Information Technology

The Superior University, Lahore

Fall 2026

Type (Nature of project)	[] **D**evelopment	[] **R**esearch	[] **R** & **D**
Area of specialization			
FYP ID			

Project Group Members

Sr.#	Reg. #	Student Name	Email ID	*Signature
(i)				
(ii)				
(iii)				

*The candidates confirm that the work submitted is their own and appropriate credit has been given where reference has been made to work of others

Plagiarism Free Certificate

This is to certify that, I Your-Name Son/Daughter of Parent-Name, group leader of FYP under registration no FULL-FYP-ID at Software Engineering Department, The Superior University, Lahore. I declare that my FYP report is checked by my supervisor.

Date: DD/MM/YYYY	Group Leader:	Signature:

Name of Supervisor:	Co-Supervisor:
Designation:	Designation:
Signature:	Signature:

HoD: Dr. Muhammad Aqeel
Signature:

APPROVAL

Project Supervisor

Comments:

Name:
Date: Signature:

Project MANAGER

Comments:

Name:
Date: Signature:

HEAD of Department

Comments:

Name:
Date: Signature:

Dedication

This work is dedicated to my . . . . . .

Acknowledgements

I am really thankful to my supervisor who has . . . . . .

Executive Summary

[12 pt, Times New Roman, Justified]

[Provide a concise one-page overview of the entire project. Include: the problem addressed, the AI/Data Science approach used, the dataset, key results and performance metrics, and the conclusion. The Executive Summary must be self-contained — it should NOT reference other sections of the document and should NOT include citations.]

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List of Abbreviations

Abbreviation	Full Form
AI	Artificial Intelligence
ML	Machine Learning
DL	Deep Learning
NLP	Natural Language Processing
CV	Computer Vision
CNN	Convolutional Neural Network
RNN	Recurrent Neural Network
LSTM	Long Short-Term Memory
GAN	Generative Adversarial Network
LLM	Large Language Model

Abbreviation	Full Form
EDA	Exploratory Data Analysis
FYP	Final Year Project
API	Application Programming Interface
GPU	Graphics Processing Unit
AUC	Area Under the Curve
ROC	Receiver Operating Characteristic
RMSE	Root Mean Square Error
MAE	Mean Absolute Error
F1	F1-Score (Harmonic Mean of Precision & Recall)
mAP	Mean Average Precision
IoU	Intersection over Union
BLEU	Bilingual Evaluation Understudy
ROUGE	Recall-Oriented Understudy for Gisting Evaluation

Abstract

Include a summary of the problem statement, challenges, proposed solution, approaches, scope, and comparison with existing systems/evaluation methods, conclusion, and future directions. Recommend length is 1 page maximum. Abstract must be self-explanatory and should not include any references or shorthand notations in the abstract.

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Chapter 1: Introduction

1.1 Background and Context of Smart Poultry Farming

The poultry industry plays a critical role in the global agricultural economy and food security by supplying high-quality protein in the form of meat and eggs. Traditionally, poultry farm management relies heavily on manual inspections and human observation to monitor environmental conditions—such as temperature, humidity, ammonia levels, and ventilation—as well as the health and behavior of the flock. However, manual monitoring is prone to human error, labor-intensive, and often too slow to prevent sudden outbreaks of disease or environmental distress.

With the rapid advancement of modern technology, traditional farming is transitioning toward smart agriculture. By integrating Internet of Things (IoT) sensors, automated monitoring systems, and data-driven intelligence, modern poultry farms can maintain optimal housing conditions 24/7. This transition allows

farmers to shift from reactive firefighting to proactive management, ensuring higher yield, reduced mortality rates, and improved animal welfare.

1.1.1 Current Challenges in Poultry Management

Despite advancements in agricultural automation, many commercial and small-to-medium-scale poultry farms still suffer from significant economic losses due to inadequate environmental control and delayed disease detection. Specifically, current operational challenges include sudden temperature fluctuations, delayed hazard detection, and a lack of centralized tracking.

1.1.1.1 Operational Limitations of Manual Systems

Manual checks are typically conducted only a few times a day, meaning critical issues like ventilation failures, water supply disruptions, or early signs of respiratory illness often go unnoticed until significant damage has occurred. This project caters directly to commercial poultry farm owners, farm managers, and agricultural operators who require an affordable, automated, and real-time solution to safeguard their flocks.

1.2 Project Aim, Objectives, and Scope

The main aim of the Smart Poultry Management System is to develop an integrated poultry monitoring solution that combines Internet of Things (IoT) technology, artificial intelligence (AI), and computer vision to support environmental monitoring and the identification of potential poultry health risks[cite: 1]. The proposed system will collect environmental data through sensors, analyze relevant information, examine poultry images for predefined visual indicators, and notify authorized users when abnormal conditions or potential risks are identified. By presenting monitoring information and analysis results through a centralized interface, the system aims to support timely decision-making and improve the efficiency of poultry farm monitoring.

1.2.1 Project Objectives

The specific objectives of the Smart Poultry Management System are as follows:

1. To develop an IoT-based mechanism for collecting environmental parameters, including temperature, humidity, and ammonia levels in poultry farming environments.
2. To monitor and record environmental readings and identify abnormal conditions using defined thresholds or appropriate reference ranges.
3. To investigate and implement an AI-based approach for assessing potential poultry health risks using relevant environmental data and other suitable

data sources.

4. To integrate a computer vision component for analyzing poultry images and identifying predefined visual indicators associated with potential health concerns.
5. To develop an alert and notification mechanism that informs authorized users about abnormal environmental conditions and relevant risk assessment results.
6. To provide a centralized interface through which authorized users can access environmental readings, historical records, alerts, and available AI and computer vision results.
7. To evaluate the functionality and performance of the developed system through testing of sensor data collection, abnormal-condition detection, computer vision predictions, and notification delivery.

1.2.2 Project Scope

The scope of the Smart Poultry Management System covers the design, development, integration, and evaluation of a technology-assisted solution for poultry farm monitoring[cite: 1]. The proposed system combines IoT-based environmental data collection, AI-based risk assessment, computer vision, alert and notification functionality, and a centralized user interface. Its primary purpose is to assist poultry farm owners and managers in monitoring environmental conditions and reviewing potential health-related concerns.

The environmental monitoring component will focus on parameters such as temperature, humidity, and ammonia levels. Selected sensors will collect these readings for transmission, recording, visualization, and analysis within the system. Additional environmental parameters, such as light intensity, may be included depending on the final hardware configuration and approved project requirements.

The AI component will analyze suitable data to identify potential risks or abnormal patterns relevant to poultry farm monitoring. The computer vision component will process poultry images using an appropriate dataset and trained model to identify predefined visual indicators associated with potential health concerns. The capabilities of these components will depend on data availability, model selection, and evaluation results. Their outputs are intended to support monitoring and decision-making rather than replace professional veterinary assessment.

The alert and notification component will inform authorized users when configured environmental thresholds are exceeded or relevant risk findings are generated. The centralized interface will provide access to available sensor readings, historical information, notifications, and analysis results. The final interface features and access methods will depend on the implemented system design.

The intended users include poultry farm owners, farm managers, and other authorized personnel responsible for monitoring farm conditions[cite: 1]. The project will be developed and evaluated as an academic prototype within the available time, hardware, software, and data resources.

The project is subject to limitations associated with sensor reliability, dataset quality, model performance, and network connectivity. The system will not be considered a universal poultry disease diagnostic solution and cannot guarantee the prevention of disease outbreaks or the elimination of poultry mortality. Automated control of farm equipment will only be included if it is implemented and evaluated as part of the final system.

1.3 Report Organization

This report is organized into six chapters, each addressing a different aspect of the Smart Poultry Management System[cite: 1]:

- **Chapter 1 (Introduction):** Presents the background of smart poultry farming, identifies the problem and unmet needs addressed by the project, and describes the project aim, objectives, scope, and organization of the report[cite: 1].
- **Chapter 2 (Literature Review):** Examines existing research and systems related to poultry monitoring, IoT technologies, artificial intelligence, machine learning, computer vision, and poultry health management. It also discusses relevant approaches and identifies research gaps related to the proposed system[cite: 1].
- **Chapter 3 (Problem Definition and Methodology):** Defines the problem addressed by the project and describes the methodology adopted for system development, including relevant data collection, processing, analysis, and evaluation approaches[cite: 1].
- **Chapter 4 (Design and Architecture):** Presents the proposed system architecture and explains the relationships between hardware components, software modules, data processing components, AI and computer vision functionality, and the user interface[cite: 1].
- **Chapter 5 (Implementation):** Describes the technologies, tools, and methods used to implement the system components and integrate them into the proposed solution[cite: 1].
- **Chapter 6 (Result and Discussion):** Presents and discusses the results obtained through system testing and model evaluation. It assesses the system against its objectives, identifies limitations, and discusses possible areas for future improvement[cite: 1].

Chapter 2: Literature Review

Here's a clear guide for what goes inside a Literature Review for an AI/Data Science thesis:

Structure & Content of Literature Review (AI/DS Thesis)

Introduction to the Research Domain

Give a broad overview of your specific field. For example:

- If your project is on **NLP** → discuss how language models evolved from rule-based → statistical → deep learning → transformers
- If it is **Computer Vision** → from hand-crafted features → CNNs → ViT
- If it is **Predictive Analytics** → from classical statistics → ML → AutoML

Tell the reader: *what is this field, how did it grow, and where does it stand today?*

Review of AI / ML Techniques

Discuss **classical machine learning methods** that have been tried for your problem type. For example:

- SVM, Naive Bayes, Logistic Regression, Decision Trees, Random Forest, XGBoost
- Explain what each does, how it was used in related papers, and what its limitations are
- You don't have to include all — only those **relevant to your problem**

Review of Deep Learning Approaches

Discuss **modern deep learning models** used for similar problems:

- For text → LSTM, BERT, GPT, RoBERTa, T5
- For images → CNN, ResNet, VGG, EfficientNet, YOLO, Vision Transformer
- For tabular data → Deep Neural Networks, TabNet
- For time series → LSTM, GRU, Temporal CNN, Transformer

For each model/paper: mention the authors, what they did, what dataset they used, what result they got, and what limitation remained.

Review of Datasets & Benchmarks

Mention the **datasets other researchers used** for similar problems:

- Name of dataset, size, number of classes, source
- What tasks it has been used for
- Why it is or isn't suitable for your work
- Then justify your own dataset choice

Existing Tools & Frameworks

Briefly mention tools and libraries used in related work:

- TensorFlow, PyTorch, Scikit-learn, HuggingFace, Keras, OpenCV, SpaCy, NLTK, FastAI
- Don't just list them — explain **why researchers chose them** and which you will use

Gap Analysis

This is the **most important part** of the literature review. After reviewing all related work, clearly answer:

- What problem is still unsolved?
- What limitations do existing methods have?
- What dataset gaps exist?
- What your project will do differently

A good format is a **comparison table**:

Table 1: Table 1 Example of Comparison Table

Author / Year	Method Used	Dataset	Results	Limitation
Smith et al. 2021	BERT	IMDb	92% Acc	Only English, small dataset
Ahmed et al. 2022	CNN-LSTM	Custom	88% F1	No real-time inference

Summary

Write 1 short paragraph summarizing:

- What the literature tells you overall
- What the dominant approach is in your field
- What gap your project fills

- How the reviewed work guided your methodology and model choice

Table 2: Table 2 Example of Table

Column 1	Column 2	Column 3	Column 4	Column 5

Chapter 3: Problem Definition and Methodology

This chapter explains the core problem that this project aims to solve and describes the step-by-step approach followed to address it. First, the problem is clearly defined in simple terms, explaining what the system is expected to do, what kind of data it works with, and what results are expected. Next, the overall research methodology is described, which includes how the data was collected, how it was cleaned and prepared for use, and how the model was trained and tested. The dataset used in this project is also described in detail, including its source, size, and structure. Basic data analysis was performed to better understand the data before feeding it into the model. This chapter serves as the backbone of the project, as all decisions related to data and model development are explained here in a clear and organized manner.



Fig. 1 [Caption of figure]

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Chapter 4: Design and Architecture

This chapter presents the overall design and architectural structure of the proposed AI/Data Science system. It explains how the different components of the system are organized and how they work together to solve the problem defined in the previous chapter. The chapter begins with a high-level overview of

the system architecture, showing how data flows from input through processing to final output. It then describes the AI or machine learning model selected for this project, including the reasons for choosing it over other available options. The training strategy, hyperparameter settings, and the end-to-end pipeline of the system are also explained in a clear and structured manner. Additionally, the tools, libraries, and frameworks used during development are listed and briefly justified. Where applicable, the deployment setup of the system is also discussed. Overall, this chapter provides a complete technical blueprint of the system, giving the reader a clear understanding of how the proposed solution was designed and built.

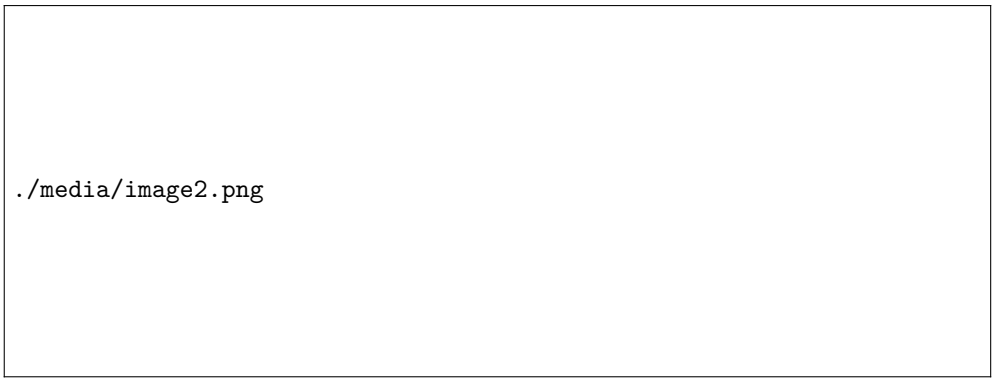


Fig. 2 [Caption of figure]

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Chapter 5: Implementation

This chapter describes the practical implementation of the proposed AI/Data Science system based on the design and architecture discussed in the previous chapter. It explains how the different components of the system were developed and integrated to create the final working solution. The chapter begins with the development environment, including the programming language, tools, libraries, and frameworks used during implementation. It then explains the implementation of the data preprocessing steps, feature preparation, and model development process in a clear and systematic manner. The training and testing procedures are also discussed, along with the integration of the trained model into the overall system. Where applicable, the implementation of the user interface, backend services, database, APIs, or deployment components is also presented. Important implementation details, screenshots, code outputs, and system workflows are included to demonstrate how the proposed methodology was converted into a functional system. Overall, this chapter provides a detailed explanation of how the proposed solution was implemented and how its individual components work together.

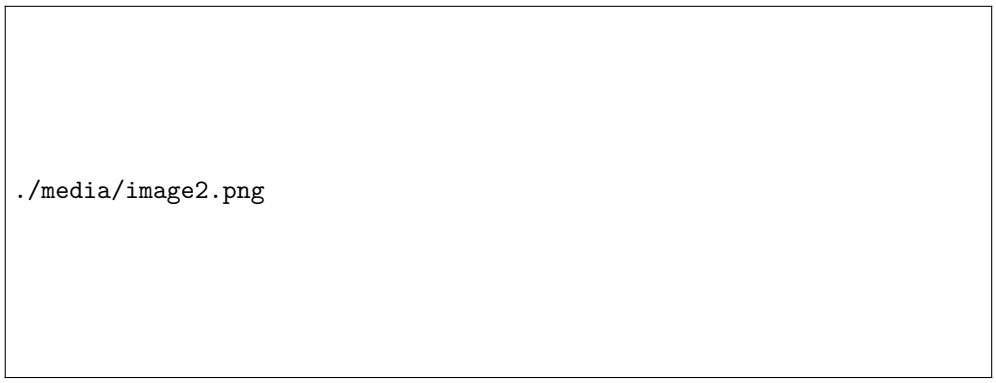


Fig. 3 [Caption of figure]

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Chapter 6: Result and Discussion

This chapter presents and discusses the results obtained from the implementation and evaluation of the proposed AI/Data Science system. It begins by presenting the performance of the developed model using appropriate evaluation metrics and testing results. The results are analyzed to determine how effectively the system solves the problem defined in the earlier chapters. Where applicable, tables, graphs, charts, confusion matrices, or other visual representations are used to provide a clear comparison of the model's performance. The chapter also discusses the observations obtained during testing, including the strengths, limitations, and possible sources of errors in the system. If multiple models or approaches were evaluated, their results are compared to identify the most suitable solution. The practical performance of the complete system is also discussed based on its accuracy, reliability, efficiency, and usability. Finally, the findings are interpreted in relation to the project's objectives, explaining whether the proposed system successfully achieved its intended goals. Overall, this chapter provides a clear evaluation of the developed solution and establishes the significance of the results obtained.

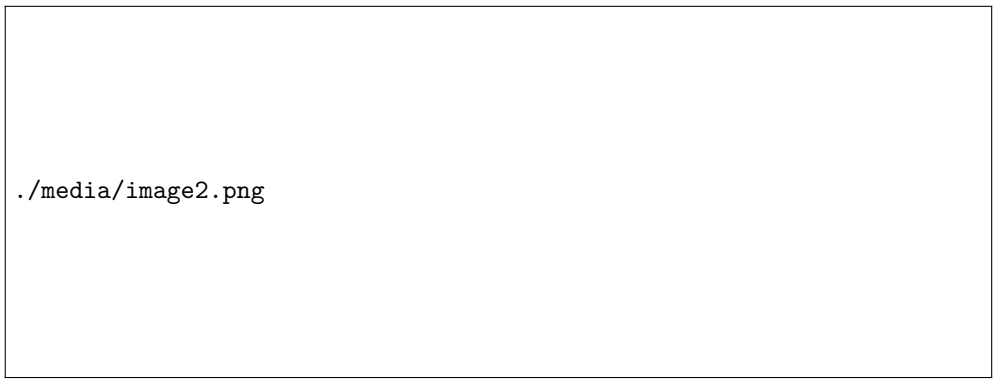


Fig. 4 [Caption of figure]

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[1] M. Sher, M. Rehman, "*Title of the Paper*" Conference name/Journal Name, Edition, Volume, Issue, ISBN/ISSN, PP, Publisher/City-Country, Year.

[2] M. Sher, M. Rehman, "*Title of the Paper*" Conference name/Journal Name, Edition, Volume, Issue, ISBN/ISSN, PP, Publisher/City-Country, Year.

[3] M. Sher, M. Rehman, "*Title of the Paper*" Conference name/Journal Name, Edition, Volume, Issue, ISBN/ISSN, PP, Publisher/City-Country, Year.

[4] M. Sher, M. Rehman, "*Title of the Paper*" Conference name/Journal Name, Edition, Volume, Issue, ISBN/ISSN, PP, Publisher/City-Country, Year.